Air-Gapped Predictive Copilot for Secure MPLS Operations

Faculty rubric compliance matrix — generated 2026-10-19

Evidence-based estimate. Features that only appear in documentation are not awarded marks.

Category	Max	Estimated
1. Problem Statement & Project Planning	2	1.5
2. Database Design	6	6
3. Database Implementation	8	8
4. Application Development	4	4
5. Authentication, Authorization & Database Security	5	5
6. Professional Development Practices	5	1.5
Total	**30**	**26.0**

Estimated bonus: +1 (automated testing + audit logging verified). CI/CD written but not run; no custom domain.

1. Problem Statement & Project Planning — 1.5/2

Requirement	Implementation	Location	Test	Result	Status	Missing work
Project title, problem statement, objectives, scope	Written proposal	docs/01_project_proposal.md	Inspected	Present	PASS	—
Team members & responsibilities	Editable template only	docs/01_project_proposal.md	Inspected	Placeholders	PARTIAL	Fill real names/IDs and tasks
Technology stack & deliverables	Table incl. documented CRA deviation	docs/01_project_proposal.md	Inspected	Present	PASS	—
Gantt chart	Mermaid gantt + week table (template dates)	docs/01_project_proposal.md	Inspected	Present	PASS	Adjust dates to semester

2. Database Design — 6/6

Requirement	Implementation	Location	Test	Result	Status	Missing work
ER diagram	Generated from ORM metadata: detailed + presentation (PNG/SVG/PDF) + Mermaid	docs/er/	Generator run; 13 tables / 24 FKs	Matches schema	PASS	—
Min. 4 primary entities	13 tables	backend/app/models.py, database/schema.sql	pytest test_connection_and_tables	Passed	PASS	—
Min. 3 relationships	24 foreign keys (1:N, 1:1)	docs/07_08_er_diagrams.md	pytest test_foreign_keys_and_indexes	Passed	PASS	—
Relational schema	Generated from metadata	docs/09_relational_schema.md	Inspected	Present	PASS	—
Keys & constraints (PK, FK, UNIQUE, CHECK, NOT NULL, DEFAULT)	23 CHECK constraints, UNIQUE keys, FK actions	database/schema.sql	pytest constraint tests	Passed	PASS	—
Normalization 3NF	Analysis with justified exceptions	docs/11_normalization_3nf.md	Inspected	Present	PASS	—
Data dictionary (recommended)	Generated per table	docs/10_data_dictionary.md	Inspected	Present	PASS	—

3. Database Implementation — 8/8

Requirement	Implementation	Location	Test	Result	Status	Missing work
Database creation & table design	Alembic 0001_baseline + schema.sql	backend/alembic, database/schema.sql	pytest schema tests	Passed	PASS	—
CRUD	ORM create/read/update/delete via API	backend/app/routers/devices.py etc.	pytest test_device_crud_and_validation	Passed	PASS	—
JOIN queries	Reports, Database Lab, ORM joins	app/routers/dblab.py, app/reports.py	pytest test_join_and_aggregate_queries	Passed	PASS	—
Views	v_device_health, v_prediction_report	database/views.sql	pytest test_views_return_rows	Passed	PASS	—
Aggregate queries	Dashboard GROUP BY, DB Lab, procedure	app/routers/reports.py	pytest + E2E dashboard	Passed	PASS	—
Transactions (commit & rollback)	Prediction workflow, actions, transaction demo, procedure	app/routers/predictions.py, dblab.py	pytest test_transaction_demo_rollback_and_commit	Passed	PASS	—
Stored procedure / trigger	2 procedures, 5 triggers	database/procedures.sql, triggers.sql	pytest procedure + trigger tests	Passed	PASS	—
ORM integration	SQLAlchemy 2 for all app access	backend/app/	pytest API tests persist to MySQL	Passed	PASS	—

4. Application Development — 4/4

Requirement	Implementation	Location	Test	Result	Status	Missing work
Framework-based application	FastAPI + React	backend/, frontend/	Build + E2E	Passed	PASS	—
Dashboard	Stats, charts, recent activity	frontend/src/pages/Dashboard.js	E2E	Passed	PASS	—
Search / filtering / pagination	All list pages	frontend/src/pages/*	pytest test_pagination + E2E	Passed	PASS	—
Validation	Pydantic + frontend validators + DB CHECK	app/schemas.py, src/lib/validation.js	pytest + Jest 4 tests	Passed	PASS	—
Exception handling	Mapped 401/403/404/409/422/429/500	backend/server.py	pytest	Passed	PASS	—
Responsive UI	CSS breakpoints 1100/860 px	frontend/src/App.css	E2E 390 px + screenshots	Passed after fix	PASS	—

5. Authentication, Authorization & Database Security — 5/5

Requirement	Implementation	Location	Test	Result	Status	Missing work
JWT authentication	Login, me, logout with revocation, expiry	app/routers/auth.py, app/security.py	pytest auth tests + E2E	Passed	PASS	—
RBAC using single Users table	users + roles; require_roles on every endpoint; engineer scoping	app/security.py	pytest test_rbac_forbidden_actions	Passed	PASS	—
ORM-based authorization	scope_to_assigned() filters SQLAlchemy queries	app/security.py	pytest test_engineer_sees_only_assigned_devices	Passed	PASS	—
Password hashing	bcrypt cost 12	app/security.py	pytest test_password_is_bcrypt_hashed	Passed	PASS	—
SQL injection prevention	ORM + bound parameters, whitelisted filters	app/reports.py	Code inspection	No string-built user SQL	PASS	—
Secrets management	.env + .env.example + .gitignore	backend/.env (excluded), .env.example	Inspected ZIP	No secrets packaged	PASS	—

6. Professional Development Practices — 1.5/5

Requirement	Implementation	Location	Test	Result	Status	Missing work
GitHub: progressive commits & branching	Documented main/dev workflow	README.md §10	—	No repository exists	NOT VERIFIABLE	Create repo; commit progressively; create dev branch
.gitignore	Excludes .env, venvs, node_modules, DB data	.gitignore	Inspected	Present	PASS	—
Database backup & restore	scripts/backup.sh, guide	docs/17_backup_recovery.md	Dump + restore into test DB	Passed	PASS	—
Docker	docker-compose.yml, 2 Dockerfiles	docker-compose.yml	Not executed (no Docker in env)	Unverified	PARTIAL	Run docker compose up and record evidence
Docker Hub	Instructions only	README.md §11	—	Not done	MISSING	Push images to Docker Hub
Cloud deployment (live URL)	Guide only	README.md §11	—	Not deployed	MISSING	Deploy with a MySQL-capable host and record URL

Bonus opportunities

Bonus	Implementation	Location	Status	Estimate
CI/CD pipeline (GitHub Actions)	Workflow written (pytest on MySQL + frontend build)	.github/workflows/ci.yml	PARTIAL — not run	+0 until it runs green on GitHub
Custom domain	None	—	MISSING	+0
Additional best practice: automated testing + append-only audit logging	37 pytest + 4 Jest + E2E; trigger-enforced audit	tests/, database/triggers.sql	PASS	+1